

Victus Exergame: An Approach to Rehabilitation of Amputees Based in Serious Game

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Abstract: This article describes a solution that aims to make the process of physical rehabilitation more attractive for amputees through a solution based on medical informatics and gamification. Considering the patient's difficulties during physiotherapy sessions (such as trauma, the pain recovery process, and motivation), this work proposes a serious game using sensors and microcontrollers that allow more engaging and challenging sessions. This relaxed approach is intended to direct the patient to a state of immersion in treatment, resulting in greater effectiveness in physiotherapy, and provide a tool for the therapist to monitor the evolution of patients during this process. Some initial experiments were performed, and the results showed that the tool is promising in proposing a more playful and motivating physical rehabilitation environment.

Keywords: Physical Rehabilitation, Embedded Systems, Serious Game, Exergame

1 Introduction

According to CIEZA *et al.* [2020], there are 180 million people around the world who have some type of amputation, whether caused by congenital malformation, trauma, vascular diseases, cancer, serious infections, or complications of chronic diseases. Amputation accounts for 52% of years lived with disability worldwide. Although the amputation mortality rate decreased, the prevalence of the condition increased by 36% between 1990 and 2019. These data highlight the importance of physical rehabilitation to improve the quality of life of people with amputation and reduce its impact on the global burden of diseases [CIEZA *et al.*, 2020].

Within the field of physical rehabilitation of amputees, during physiotherapy, patients need to overcome several obstacles to perform a successful treatment. Due to the proximity in which the amputation occurred, the pain during the rehabilitation process, the lack of acceptance of the event, and even phantom pains of the amputated limb are some challenges faced by the patient and, consequently, by professionals in the area in aid of these patients. As much as it is a time to get around the patient's difficulties, it is not always easy to keep him motivated to engage with the adaptive treatment MATOS *et al.* [2014]. In this sense, technology has been a great ally in helping this process, primarily through gamification.

Even with the advancement of technology, there are still few low-cost solutions to assist physical rehabilitation. On the other hand, the leading solutions to make the physiotherapy process more attractive involve hardware devices with peripherals capable of capturing information and user movements, such as video game consoles. However, performing

an analysis based on the patient's vital signs is impossible through these devices. Due to this lack of data, the analysis of the physiotherapist about the evolution presented by the patient is often subjective because it is supported by human observation and patient descriptions [MOUSAVI HONDORI and KHADEMI, 2014].

In this context, this work proposes a system to transform physical rehabilitation sessions into a more playful and motivating activity through a serious game. Focusing on patients with lower limb amputation, who undergo exercises on a stationary bike to adapt to their prosthesis. Due to the demand observed by the Physical Rehabilitation Service of Bagé (SRF), activity is monitored using sensors to capture physiological data, bringing a greater detail of the patient's evolution from the aid in its evaluation by the physiotherapist in the different sessions. The following sensors are used in this solution: heart rate sensor, muscle strength sensor, and Hall effect sensor (which, together with a magnet fixed on the wheel of the exercise bike, allows measuring speed and displacement during the session).

The first version of the system was developed. After some initial experiments, a new version was proposed, considering new challenges and points system and making available a new phase with a new scenario and architectural modifications. The new architecture used a new system based on web application for presenting the data sessions. This way, from databases developed on the Firebase, it is possible to present the results for the physiotherapists with a more suitable tool. Thus, after new functional experiments were performed, it was possible to verify that the new solution was more interesting for physiotherapists and patients than the initial version. This work is organized as follows: Section 2 provides

the theoretical basis on the subject of the project; Section 3 describes the exploration of the project space and presents the developed solution; in Section 4 are presented the results of the experiments performed; finally, Section 5 contains the conclusions of the work and the bibliographical references.

2 Theoretical Foundation

The amputation process is necessary in cases of accident or illness in which there is no option to save or improve the patient's quality of life [MATOS *et al.*, 2020]. Amputation is not only about removing the limb because, after the surgical procedure, the patient also begins the process of physical rehabilitation to learn how to deal with his new condition and readapt to daily activities. This process involves several health professionals (such as physiotherapists, occupational therapists, and psychologists) to recover the patient's mobility, independence, and self-esteem. The process involves adapting to prostheses, improving balance and motor coordination, and exercises to strengthen muscles and improve mobility. Finally, post-amputation involves psychological and emotional care through therapy MATOS *et al.* [2020].

The physical rehabilitation process can be optimized through sensing devices that collect real-time patient data and send information to medical monitoring systems. Embedded systems, such as those intended for sensing, are widely used in medicine, such as vital sign monitors, diagnostic devices (electrocardiograms, electroencephalograms, and tomography), and life support devices. These devices control the patient's vital functions, such as mechanical ventilation, medication administration, and blood oxygenation monitoring, ensuring patient survival in emergencies [ARANDIA *et al.*, 2022]. The use of computational resources in medicine, such as sensors and virtual games as the solution described here, belongs to Medical Informatics. From the perspective of computer science, this is the field of study that ranges from the collection, treatment, and presentation of data to the health professional to more hardware-oriented tools, such as the development of measuring instruments and robotic prostheses, among others [GRAY and SOCKLOW, 2016].

The sensors in this system have two purposes: monitoring physiological data for a greater understanding of the state of evolution of the patient and controlling the simulation game to unwind the activity through gamification, which, despite being a separate concept, is present in this context. Gamification is a technique that uses elements of games in non-ludic environments in order to improve the involvement in solving a given problem [BERTRAN, 2014]. This approach aims to stimulate the participation of patients through the use of resources such as scores and rewards for these activities and make the treatment environment more enjoyable, in addition to allowing visualization of the progress of each of them.

2.1 Correlated work

In recent years, interest in using serious games exergames that promote physical activity) in physical rehabilitation has grown, particularly for patients with motor difficulties

[VAN DIEST *et al.*, 2013]. As in amputees, in order to make rehabilitation more engaging.

NASCIMENTO *et al.* [2023] presents a relevant solution in physical rehabilitation for patients with motor difficulties. The work presents an innovative approach to assistance in telerehabilitation sessions (domestic physical rehabilitation sessions, where the patient and physiotherapist are in different environments and interaction is mediated by communication technologies). It is demonstrated in the exergame "CicloExergame", a game genre Endless Runner (style similar to the game Subway Surfers), which aims to keep as long as possible, avoiding obstacles and collecting coins to add to the score. The system consists of a Cicloergometer, an oximeter, a numeric keypad (for controlling the lateral movement of the character in the game), and an Arduino to perform the connection and analysis of the data obtained in the Cicloergometer (number of pedals) and the oximeter (heartbeat per minute - BPM - and blood saturation - spo2) [NASCIMENTO *et al.*, 2023].

PRAHM *et al.* [2017]. Also uses an exergame, in this case, to adapt myoelectric prostheses (prostheses controlled by muscular electrical signals) in people with amputation below the shoulder. Three games were presented: the super-tuxkart racing against time and computer-controlled opponents, Step Mania 5, which is a rhythm game (genre similar to games like Guitar Hero and Just Dance) whose goal is to activate the "notes" in order and at the right time, making the virtual character dance without errors. Each game has its particularity in the controls. However, they are generally sent via an electromyography sensor (EMG) fixed in the muscles where you want to apply rehabilitation [PRAHM *et al.*, 2017]. The study results show that individuals who used the mentioned games performed better in myoelectric prosthesis control tests than those who received only traditional rehabilitation training. Although myoelectric prostheses are not the focus of this study, the participants of the tool mentioned above reported that the game was more fun and engaging than the classic exercise.

Games that promote physical activities in patients during the telerehabilitation process are also aimed. "ExerCam" is a serious game that uses the technology of "Human Pose Estimation" to detect the position of the player's body via a standard camera tracing lines highlighting the joints - the result of which resembles a stick figure drawing. Using a standard camera, the player performs interactions with the tool, can visualize on the screen, and can perform the challenges proposed by the four games available in the system. Crazy Targets is a game where players must hit the circular targets on the screen. Boxing is a game that stimulates the patient's upper body through boxing movements to hit the targets that appear on the screen. The third game is Football, in which the patient must kick a virtual ball to exercise the lower body. Finally, the last game is called Bars. Also, to exercise the lower limbs, the player has a goal to dodge bars that will appear on the screen of the patient getting up or sitting. To account for a higher score, aspects such as accuracy and speed of movements are crucial [Rosique *et al.*, 2021]. The games present in this application have varied levels, providing different challenges depending on the stage where the patient is, an interesting aspect to be based on and bring to

the present study, considering that amputee patients react in very different ways to the process of physical rehabilitation, varied challenges are essentially important in this context.

In summary, the studies show promising results about the use of serious games in physical rehabilitation activities in patients with motor difficulties, both in improving the control of myoelectric prostheses and in the second work, physical activities in sessions of distance physiotherapy (telerehabilitation sessions), offering greater engagement, immediate feedback and removing monotony during recovery. However, although the applications presented are low-cost, these works seek to reach a different target audience of the system described here (people who go through the amputation process).

3 Victus Exergames

The proposed work aims to create a low-cost system that unites an exergame to a monitoring structure for physiotherapy sessions. The patients targeted for this application are those with lower limb amputations who perform exercises on the exercise bike. Therefore, this work can be divided into two parts: the physiotherapist interface with patient data obtained by the sensors and reporting for each session. Another uses the same sensors, but their data controls the bike in the simulation game, where the patient must travel and avoid obstacles on the track of the virtual environment in order to maximize their score.

The monitoring system emerged due to the demand identified by physiotherapy professionals active in the Physical Rehabilitation Service of Bagé/ RS. It was recognized, due to the empirical analysis of the progress of patients, that there is a need for tools that assist in the professional diagnosis so that this treatment can be monitored. The progress of the patient is recorded and then evaluated.

From this, a system was developed with node sensors connected to a microcontroller that processes the collected data in a format more understandable to the physiotherapist and performs the necessary calculations to capture the heartbeat and the speed calculation achieved by the exercise bike. This data is then sent via serial communication to a computer that has a database for storing information in a structured way. This information is accessed through software developed to elaborate graphs about the session time for a more detailed analysis of the patient's performance during the exercise. Finally, through this same software, it is possible to generate documents (reports in PDF format) containing data referring to the history of sessions of a particular patient, including partial data and charts. The solution based on a serious game approach derives from the tests carried out from the first part of the project only with data capture and reporting. It was noted that the lack of tools that solve the low motivation in amputees during physical rehabilitation sessions.

Then, delimited this problem, this game adapts the information obtained and sent by the sensors to the physiotherapist monitoring software for a 3D game developed in Unity that aims to simulate a bike route to be performed by the patient, allowing a more relaxed and engaging moment. It then uses the speed of the exercise bike to control the speed of the

game, displayed on the screen along with the data obtained from beats per minute (BPM) and electromyography (EMG) in order to give a visual perception of the effort committed by the patient during the activity.

3.1 Solution Modeling

For the implementation of the system that captures information about the rehabilitation exercise, already elucidated, it was necessary to seek sensors both to adapt the stationary bike to obtain the speed achieved in real-time and to analyze the physiological data of patients. Similarly, it was also necessary to seek essential concepts for creating games and modeling the session management software, finally obtaining a solution architecture as shown in Figure 1.

In Figure 1, it is possible to see that the information about the sensors is sent to the microcontroller and then used by the game which, at the end of a session, sends the obtained data to the Firebase online database. Also, the data session is sent to the physiotherapist tool for analysis. The modeling process of the solution is presented hereafter.

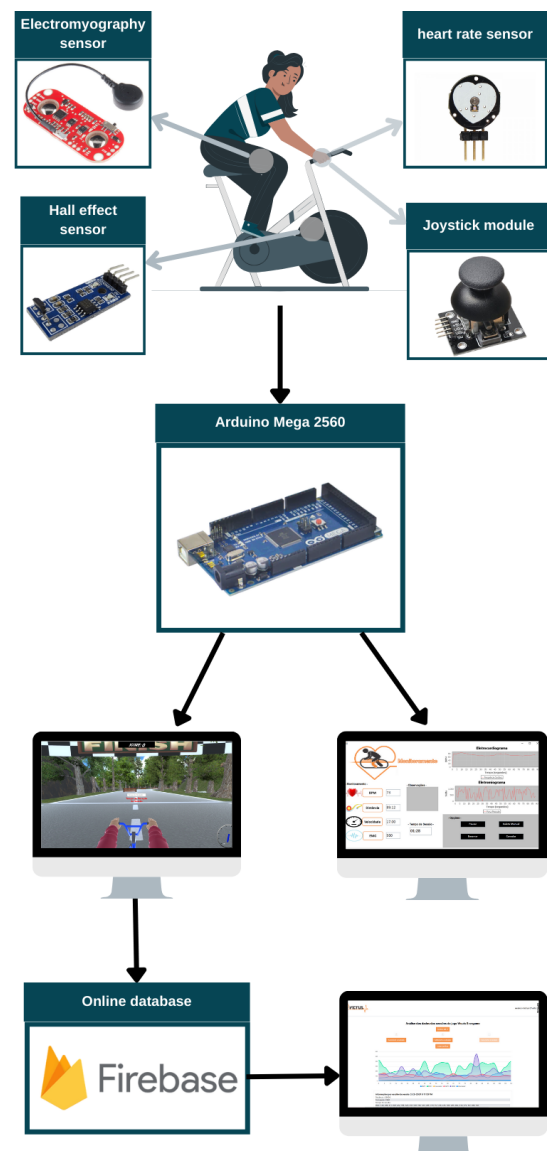


Figure 1. Conceptual Diagram of the Tool and Positioning of the Sensors.

3.2 Sensors Modeling

The modeling of the sensor node is one of the fundamental steps in developing electronic devices for monitoring physiological signals in clinical and sports environments. This subsection will present the four types of sensors used to monitor and control the game during the rehabilitation exercise. These are intermediated by the Arduino Mega 2560 hardware prototyping platform, attached to the patient's body and exercise bike: the Amped sensor, the hall effect sensor, the electromyography sensor, and the joystick module. The project's low-cost goal was to consider simpler and cheaper sensors to build the solution.

3.2.1 Pulse Amped Sensor

The Pulse Amped sensor is an electronic device for a microcontroller capable of measuring the heartbeat rate through photoplethysmography (PPG) technology [LLC., 2015]. It consists of a small module attached to the finger that emits light and measures the amount of reflected light to determine changes in blood flow, as shown in Figure 2.

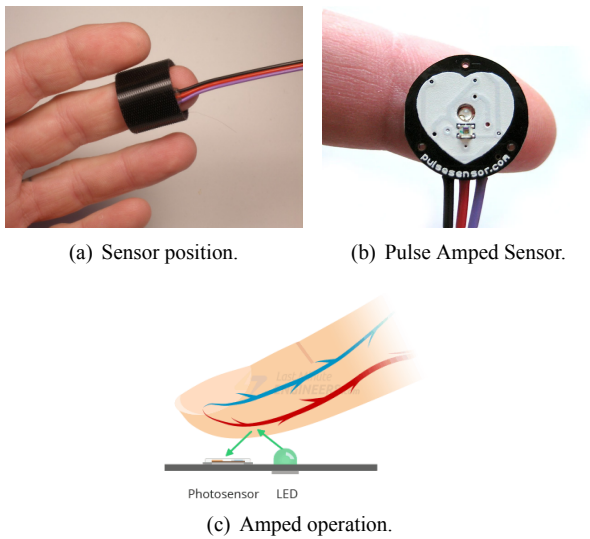


Figure 2. Pulse Amped Sensor, positioning and its operation.

3.2.2 Electromyography sensor

The MyoWare Muscle Sensor, chosen to monitor muscle strength, measures electrical signals through the skin surface that the muscle emits when it contracts or relaxes through electromyography. The sensor and its arrangement in the area to be evaluated and its mode of operation are shown in Figure 3.

This sensor uses two silver electrodes to capture the electrical signals generated by the muscle and another to serve as a ground that must be attached away from the muscle it is planned to analyze. These signals are then amplified and filtered to remove unwanted noise, and the resulting signal is emitted as an analog voltage that can be picked up by the microcontroller [TECHNOLOGIES, 2015].

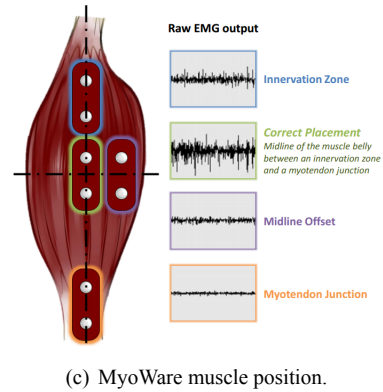
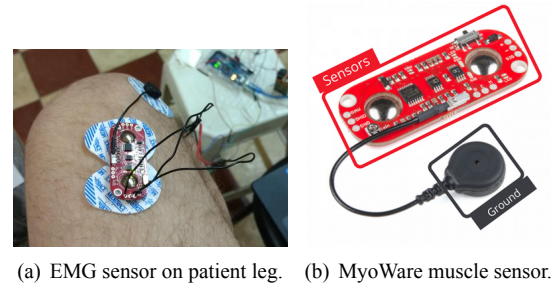


Figure 3. EMG sensor, its positioning in muscle extension and functioning.

3.2.3 Hall effect sensor

The Hall 3144e Effect sensor (Figure 4) is an electronic device that detects the presence and strength of a magnetic field through the Hall Effect principle, which defines that when a conductor carrying a current is placed in a magnetic field, a voltage is generated perpendicularly to the current and magnetic field. It consists of a module containing a thin strip of semiconductor material sensitive to changes in magnetic fields. The strength of this current is proportional to the strength of the magnetic field, which can be measured and displayed by an Arduino board connected to the computer, for example. This sensor does not measure the physiological signs of the patient but rather the number of times a magnet attached to the wheel of the exercise bike passes through the sensor, as shown in Figure 4(b), returning the wheel rotation per minute of the equipment calculated in the Arduino for use in the session management software and for the game [MICROSYSTEMS, 2022].

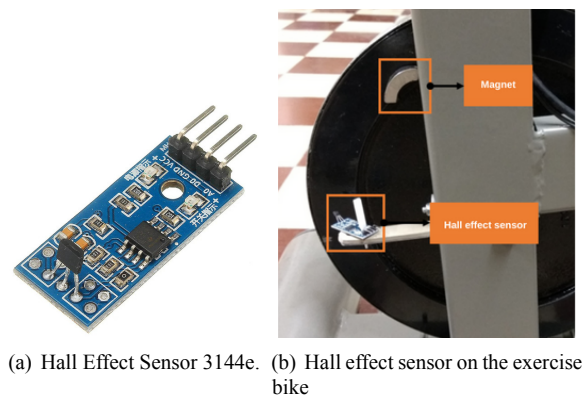


Figure 4. Hall effect sensor and its position

3.2.4 Joystick module

KY-023 is an electronic device that works with two potentiometers connected perpendicularly, one for horizontal movements and the other for vertical movements, in addition to a push button positioned at the center of the axes as illustrated in Figure 5 [JOY-IT, 2017].



Figure 5. Joystick module KY-023.

3.3 Serious Game

The game uses all the sensors mentioned above to control the app designed for the patient, making the sessions more playful. Thus, a game was developed through the Unity software in addition to the monitoring tool of the information monitored in the exercise. Whose interface is contained in Figure 6, in Figure 6(a), it is possible to see the main interface that allows two flows execution. The first selected play game, where the next step is to select the track play, time of session, and patient. On the second, which also has a menu of settings shown in Figure 6(b), to adjust the sound and difficulty of the game, it is possible to set whether the game will be controlled through the sensors or the keyboard (debug mode).



(a) Game main menu.



(b) Settings menu.

Figure 6. Game menu interface.

The game has a 3D scenario with a path to be fulfilled by the patient with visible obstacles to avoid, which will randomly spawn according to the difficulty selected by the physiotherapist on the previous menu (Figure 6(b)). The functions of accelerating the virtual bike are in charge of the Hall effect sensor, which, together with a magnet positioned on the bicycle wheel, allows calculating the speed of the patient in the game, which makes it possible to inform the distance traveled on the exercise bike.

The challenges in the game are shaping as the patient evolves, and these challenges are visible barriers ahead on the track to be avoided through the joystick module fixed on the bike's handlebars. With each barrier bypassed, the player adds points proportionally to their speed, and in case they collide with the barriers, the points will not be computed. On the game screen presented in Figure 7, it is also possible to see highlighted in different colors that the heart rate, the speed at which the bike reaches beyond the force affected by the patient's musculature (measured by the electromyography sensor) and, time session.

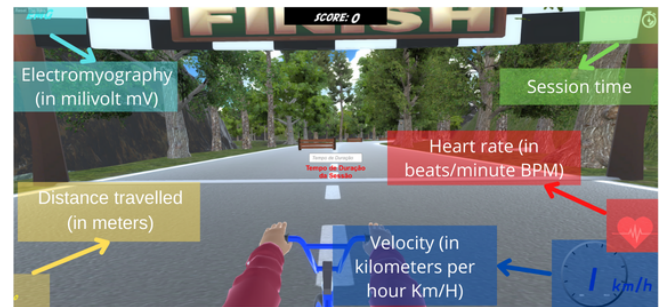
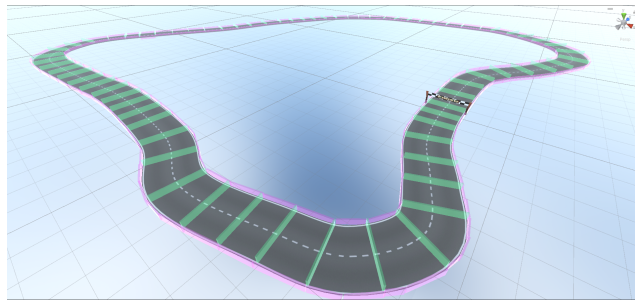


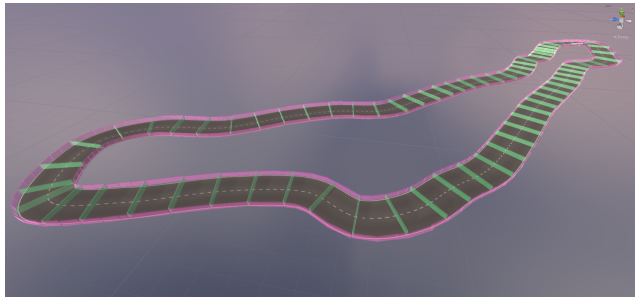
Figure 7. Game screen designed for the patient.

Besides that, considering the patient's constraints, some features to help them were built. First, a side barrier system was developed to avoid wasting time returning to the road. At game time, this invisible system was placed around the track and can be recognized when the patient is off the road. Suppose the patient goes to the wrong side of the track. In that case, the game has a system of waypoints to avoid the incorrect path formed by invisible objects during the track to identify collisions, causing the player to return to the last Waypoint after 2 seconds of the detour. When it happens, the bike will be transported to the center of the road. This way, the patients will not be demotivated during the learning curve. After a time, the system can be turned off on challenging levels. This system of waypoints developed in Unity can be shown in Figure 8 (a, and b).

Furthermore, the game features two distinct tracks, each offering a unique experience to the player. One of the tracks takes place in a forest setting, while the other takes the player on a pedal ride by the sea (exclusively developed for this work), with a sunny climate, a lighthouse, and the sound of breaking waves. Both tracks are designed with a clearly defined finish line, providing instant visual feedback to the patient about their progress on the track. The two tracks can be seen in Figure 8 (c, and d), highlighting the diversity of environments and stimulating different visual experiences for players escaping the monotony of stationary exercise.



(a) Track 1.



(b) Track 2.



(c) Forest track.



(d) Beach track.

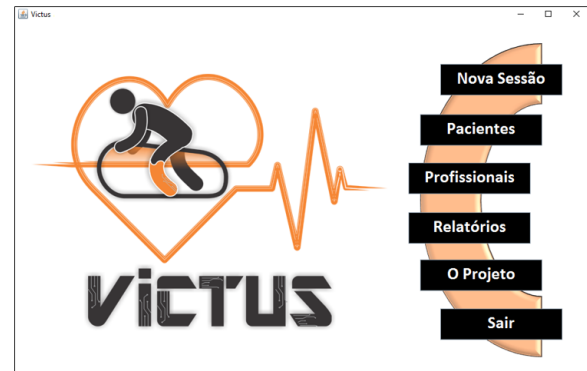
Figure 8. Side barrier and waypoints system, and Tracks available in the game.

3.4 Tool for Physiotherapists Reports

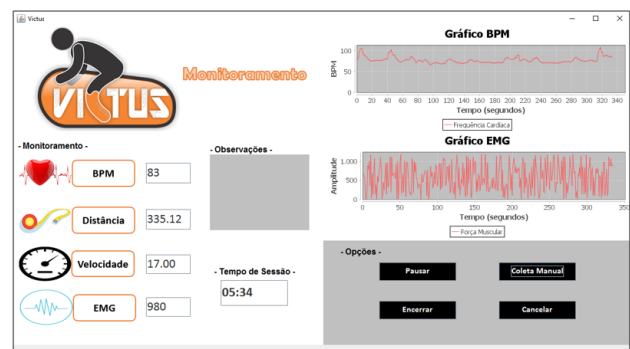
All sensors are connected to the Arduino Mega 2650, which is responsible for collecting data, processing it, and sending it via serial communication to a database on the computer. The data received from the session can be viewed using software that displays the information through a graphical interface developed in the Java language. In Figure 9, it is possible to observe the interface of the Java language physiotherapist's tool. Figure 9(b) shows the data obtained in real-time are presented on the monitoring screen, both numerically and with present data on BPMs, distance covered, speed, and force performed in millivolts (corresponding to 1/1000 of a volt

or mV , obtained through EMG), as well as in a graph that demonstrates the variation of values over the session time.

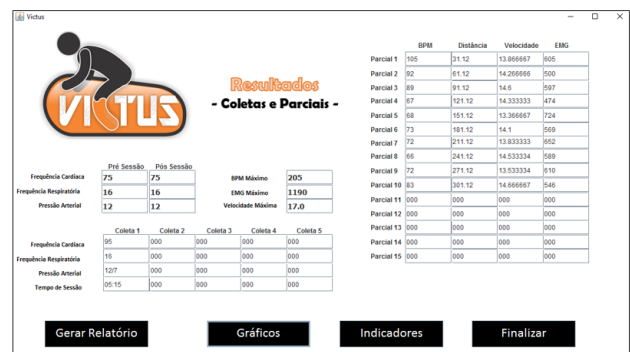
At the end of the session, the physiotherapist has some options about what to do with the data obtained, as shown in Figure 9(c). There can be a more detailed analysis of the partial results obtained during the session, a graphical analysis of the information obtained, or a report containing all partial data and the addition of physiological data obtained before the session, helping the professional to make decisions about patient advancement.



(a) Main menu screen.



(b) Real time info and charts interface.



(c) Partial results at finish of session.

Figure 9. Monitoring tool in Java and its views.

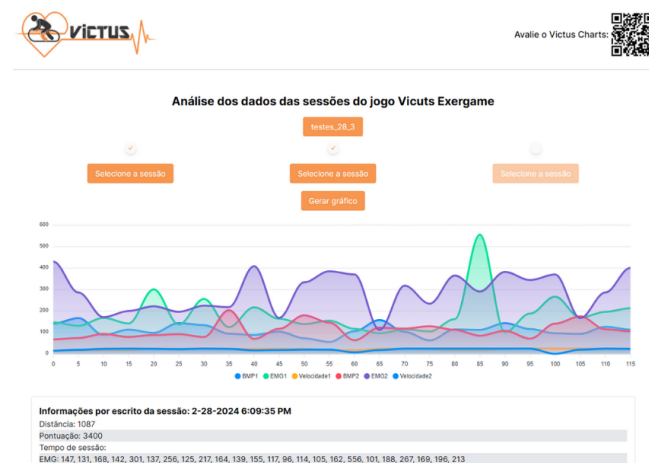
3.4.1 New Tool For Sessions Reports

As much as the approach with Java language is still valid and functional, due to problems such as the difficulty in bringing new features, global access of a tool, scalability and portability, it was decided to take a different direction in the application of physiotherapists use, a web interface. The new

approach takes the game data to an online database that can be accessed and viewed from anywhere with more dynamism and latest features for manipulation and presentation of this data.

This web interface was created and linked with the game a database in Google Firebase, a free solution for small projects (up to 1 Gb of stored data). Then, at the end of each session, all the data obtained during the exercise time defined by the physiotherapist are saved with 5 second intervals and are sent to Firebase to be accessed through a web interface. The web interface screen is presented in Figure 10.

Despite being an initial system (still being developed), a great advantage of the web tool is the dynamic visualization of the graphs, which allows comparisons between specific information from one or more sessions. An example is shown in Figure 10(a), where two sessions present every sensor's data. In Figure 10(b) it was select a comparison between the speed performed over time of the same sessions. This way, the web interface allows more portability and a friendly environment for physiotherapists; it is also possible to easily create new indicators and different reports on the system.



(a) Comparison of data obtained in two sessions.



(b) Speed-only comparison case between two sessions.

Figure 10. Charts generated from sessions data.

4 Results and Discussion

After completing implementations, both versions of the Vicus System performed experiments for all its functionalities. This experiment aims to elucidate the performance of the system in registrations, session recording, session analysis, and reports. Patients and physiotherapists were initially registered to test the solution's operation and perform session recording tests. The results happened as expected. All tests involved physiotherapy professionals who helped develop the solution prototype. In the current stage of the research, the experiments were performed in physiotherapy sessions through a partnership with the SRF of the city of Bagé-RS, which added to the project knowledge of the area of physiotherapy as well as provided the appropriate facilities for the tests.

The experiments will be reported as follows, performed in three moments. The first experiment evaluated the Vicus System without the patient's game, only the sensors and the Java Language Physiotherapists tool. The second was performed to evaluate improvements in the Vicus Exergames with only one track (forest). The third experiment was Vicus Exergames with modifications described, especially for this work, like a new track (beach), challenges and waypoints system, and a web application for physiotherapists.

The first experiment was divided into two stages, considering the tool has two independent software. The first stage was the evaluation of the accuracy of the sensor node, which was done by comparing each hardware component with the capture of manual data. In the case of the Amped pulse sensor, the captured values were compared with the readings of a digital oximeter. For the Hall sensor, the speed values obtained were compared with the speed displayed on the exercise bike display. At this stage of the EMG sensor experiments, MyoWare was not tested, as there was no base system for evaluating sensor accuracy. These individual tests of the sensors showed an equivalent operation to the reference devices and the ability to perform multiple collections in real-time, as shown in Table 1 that compares the accuracy averages per minute of the values measured by each piece of equipment. In the first minute, there was a low accuracy of the Amped sensor, which can be explained by its delay in stabilizing the reading of the luminosity signal, exposed by the sensor's documentation.

Table 1. Accuracy of sensor node.

Time	BPM	Distance	Velocity
1 st Minute	46.97%	98.60%	93.55%
2 nd Minute	84.18%	96.89%	96.16%
3 rd Minute	91.28%	96.21%	97.56%
4 th Minute	96.57%	94.72%	97.29%
5 th Minute	95.84%	96.84%	97.72%
Total	82.96%	96.65%	96.45%

In the second stage, tests were performed on integrating sensors with monitoring and session management (on Java Language), already applied in physiotherapy sessions. For this stage, the experiments have the participation of three patients selected by the professionals of the SRF, who participated in the exercises in the exercise bike monitored by the system, as shown in Figure 11. After using the tool, a multiple choice questionnaire was conducted for physiotherapists to obtain feedback on the system's usability. This step returned reasonable indications about the use of the tool, as well as ideas for possible improvements.

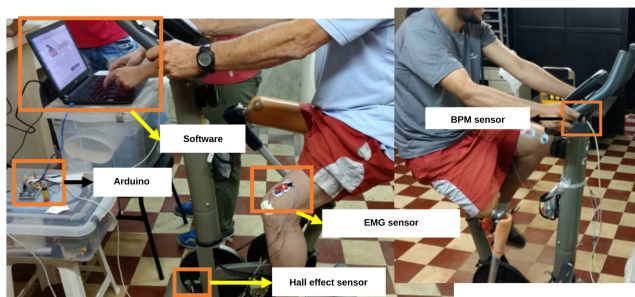
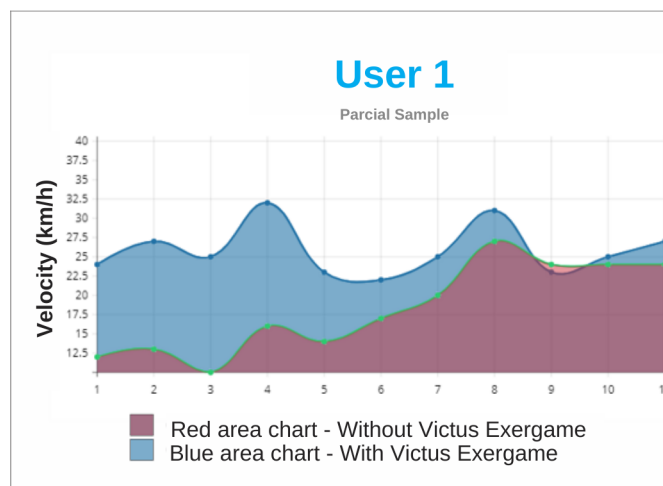


Figure 11. Test with patient system and only sensor node.

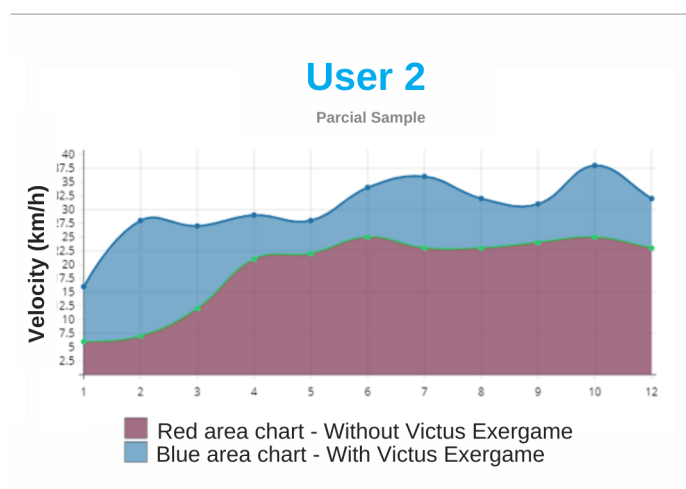
The second experiment was directed to test the game's impact during physical rehabilitation activities. The tests consisted in the use of the game during the physiotherapy session is depicted in Figure 12 still without using the modifications proposed in this work. There were two volunteer patients. In this experiment, there was also a questionnaire that, this time, was directed to both the physiotherapist and the patient based on the Likert scale to evaluate the motivational potential of the game from the perspective of both. The results on the efficiency and applicability of the tool in the real environment returned an average of results close to the maximum (grade 5). In addition, the partial speeds obtained with and without using the game were compared. With this, it was possible to observe a significant increase in speed when using the game, demonstrating greater patient involvement in rehabilitation. Figure 13 shows the chart of the comparative speeds of the sessions.



Figure 12. Patient test with the game.



(a) Graph of partial samples of speed user 1.



(b) Graph of partial samples of speed of user 2.

Figure 13. Graph of partial samples of speed.

Finally, the third experiment was directed to initial tests of modification proposed in this work, like a new track (beach), challenges and waypoints system, and a web application for physiotherapists. The tests consisted of the use of the game during the physiotherapy session. There were also two volunteer patients; in the same way, there was also a questionnaire that, this time, was directed to both physiotherapists on the Likert scale to evaluate the motivational potential of the game from the perspective of both. The results on the efficiency and applicability of the tool in the real environment returned an average of results close to the maximum (grade 5). Figure 14 presents the patients in the third experiment. In this experiment, it is essential to note that report that physiotherapists, by the motivation of patients, one this patients was ion the second session with prosthesis was able to player the game (Figure 14(b)), that demonstrate the high applicability of Victus Exergames. Also, the web application presented was considered more user-friendly by physiotherapists. Some interesting feedback was received, such as *"I suggest including a more sound stimulus in the game, which reflects the track the patient is traveling on, with a different sound when he bumps into something or leaves the track"*. In the following versions, these improvements can be made.



(a) Patient 1



(b) Patient 2

Figure 14. Patients testing Victus Exergames last version.

Despite the promising results, it is essential to emphasize that this work still aims at several improvements for the next tests. Some of them: the assignment of steering controls by adapting the handlebars of the exercise bike and connecting the distant sensors wirelessly due to the difficulty of adjusting all sensors before the sessions and the ease of unplugging a jumper.

5 Conclusions

This work proposes a monitoring system for physiotherapy sessions on the exercise bike for amputees based on gamification. A sensor node was implemented that could measure physiological and exercise data to monitor patients' progress during treatment. Several experiments were performed, and the results demonstrated that it was possible to obtain a more significant commitment from the patients with the rehabilitation sessions, thus reducing the time of adaptation and demotivation in this stage of physiotherapy. In future work, refine the physiotherapist's application and migrate to progressive web application technology that is more suitable for any device (web or mobile). Besides that, it is considered to add a new way to control the direction of the virtual bike using a controllable handlebar to increase player immersion and especially the expansion of tests within a larger group of patients with a follow-up that evolves over the sessions with the use of the game.

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